

Toqa Gamal Mohamed

Email: toqagamal611@gmail.com | GitHub : github.com/Toqa-Gamal | LinkedIn: linkedin.com/in/toqa-gamal | +20 1096735511 | New Cairo, Cairo, Egypt

PROFESSIONAL SUMMARY

Recent Computer & Systems Engineering graduate with hands-on experience in functional verification (UVM/SystemVerilog) and software testing (black-box/white-box, JUnit 5, Jira). Skilled in designing test cases using BVA, Equivalence Partitioning, and Decision Table Testing. Seeking a Junior Software Tester role to apply strong analytical and defect-detection skills.

EDUCATION

Faculty of Engineering, Ain Shams University

Class of 2026

Bachelor's degree in computer & systems engineering

Cumulative GPA: Very Good (3.1/4.0)

PROFESSIONAL EXPERIENCE

Graduation Project — Digital Verification of PCIe 5.0 Transaction Layer
Ain Shams University, Faculty of Engineering — Computer & Systems Engineering

Sep 2025 – Jun 2026

Supervised by Prof. Dr. Ayman Wahba, Ain Shams University.

Summary of Project :

- Conducted a project study of the PCIe 5.0 specification to define the functional behavior of the Transaction Layer and produced a comprehensive block-level architecture capturing the protocol's four core functionalities.
- Built UVM-based testbenches (SystemVerilog, QuestaSim) for four PCIe 5.0 TL RX modules, developing scoreboards, reference models, drivers, monitors, and functional coverage models.
- Identified and documented RTL bugs through targeted verification scenarios, including boundary/off-by-one errors, incorrect CRC complement logic, missing enable gating, and incorrect address reconstruction.
- Achieved **95%+ functional and code coverage across all verified modules**, with results documented in a comprehensive coverage report.

My Role:

Within the 8-member team, I was responsible for the functional verification of four PCIe 5.0 Transaction Layer RX modules: `tl_rx_write_handler_ecrc`, `tl_rx_error_check_unsupported_req`, `tl_rx_error_check_unexpected_cpl`, and `tl_rx_error_check_receiver_overflow`.

- Built the complete UVM verification environment for each module — including scoreboards, reference models, drivers, monitors, sequences, and functional coverage models — and validated the RTL implementation against the PCIe 5.0 specification.
- Designed test scenarios that uncovered and documented intentional RTL bugs (off-by-one boundary conditions, missing enable gating, incorrect BAR address reconstruction, and incorrect CRC complement logic), debugged compilation/simulation issues in QuestaSim, and iteratively improved functional and code coverage to 95%+ across all four modules.
- Analyzed the coverage reports and contributed to the technical write-up and presentation of the verification results.

Full Stack Web Development (PHP Track) – ITI
Final Project: Online Restaurant Management System

Jun /July 2025

- Built a full-stack restaurant ordering and reservation platform using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript.
- Designed a relational database schema (menu items, cart items, orders, order items, reservations, customers) and implemented core workflows: browsing the menu, adding items to cart, checkout, and order confirmation.
- Developed an admin dashboard for managing menu items, categories, users, and orders, including order status updates and reservation tracking.
- Implemented dynamic cart updates and form validation using JavaScript.

National Air Navigation Services Company (NANSC)

Aug 2024

- Training covering networking fundamentals, aviation data systems, and Linux administration.

Egyptian Maintenance Company (EMC)

Aug 2024

- IT training: web development basics (HTML/CSS/MySQL/JavaScript) and networking fundamentals.

North Cairo Electricity Distribution Company (NCEDC)

July /Aug 2023

- Hands-on training in transformers and low/medium-voltage grid distribution systems.

PROJECT

Testing (Movie Recommendation System) | Java, JUnit 5

Spring 2025

- Designed and implemented 350+ black-box test cases (JUnit 5) for a Java-based movie recommendation system, covering input validation, parsing, and recommendation logic.
- Applied Boundary Value Analysis, Equivalence Class Partitioning, Negative Testing, and Decision Table Testing to validate data formats and recommendation behavior against spec requirements.

MIPS 32-bit Processor | Verilog

Fall 2024

- Architected a structural 32-bit MIPS processor by connecting register files, control units, and memory blocks.
- Validated processor functionality with a custom-designed test bench.

Library Management System | MySQL

Spring 2025

- Mapped an EER diagram from ERD Plus into a relational schema and implemented the system using SQL queries by using phpMyAdmin

CPU Scheduler | C++, Qt6

Spring 2025

- Simulated FCFS, Round Robin, SJF, and Priority scheduling algorithms with real-time Gantt chart visualization using Qt.
- Achieved a dynamic and interactive scheduling interface to visualize process transitions and CPU time slices.

GPS Tracking System | Keil uVision, Tiva C

Spring 2024

- Built a real-time GPS tracker using the Tiva C microcontroller and Neo-8M GPS module.
- Displayed live coordinates and simplified hardware-software interfacing for microcontroller projects

Solar Tracker | LDRs, Motors, Op-Amps

Fall 2023

- Designed a feedback system that aligned solar panels based on LDR sensor readings
- Maximized energy efficiency by automatically adjusting the panel's position toward sunlight

Online Shopping System | Java, JavaFX

Spring 2023

- Designed an e-commerce platform with object-oriented principles and exception handling.
- Enabled browsing and purchasing with a user-friendly GUI using JavaFX

Smart Irrigation System | Sensors, Relay, DC Pump

Spring 2023

- Created an automated irrigation controller based on real-time soil moisture levels.
- Reduced water usage through conditional pump activation logic.

Solar Water Dispenser | Peltier, Solar Panel, Pump

Spring 2023

- Built a solar-powered water dispenser that delivers heated or cooled water using a Peltier module.

Simple 2-bit ALU | ICs, 7-Segments, Decoders

Fall 2022

- Constructed a functional 2-bit ALU supporting add, subtract, AND, and OR operations

COURSES

- **Software Testing Track – Digital Egypt Pioneers Initiative (DEPI) OnGoing [July 2026 / DEC 2026]**
Undergoing comprehensive training in software testing throughout the SDLC, covering ISTQB Foundation Level topics: testing fundamentals, static testing techniques, test analysis and design, and managing test activities in Agile environments.
Gaining hands-on experience with test tools including Jira, Postman, and Selenium, and building foundational Java programming skills.
Track concludes with a Capstone Project applying end-to-end software testing methodologies from planning through execution and reporting.
- **Introduction to Programming using MATLAB – Coursera**
Learn the basics of programming and problem-solving using MATLAB, with a focus on engineering and scientific applications.
- **Introduction to Java – Coursera**
Explore fundamental Java programming concepts, including syntax, control structures, and object-oriented design.

SKILLS

Programming Languages: C, C++, Python, Java, MIPS Assembly

Hardware Description Languages: Verilog, SystemVerilog

Testing Methodologies: STLC, SDLC, Bug Lifecycle, Black-Box & White-Box Testing,
Unit/Integration/System Testing, Agile environments

Verification & Testing: UVM (Universal Verification Methodology), QuestaSim, Functional/Code Coverage Analysis, JUnit 5, Jira

Software Development: Object-Oriented Programming (OOP), Data Structures and Algorithms

Embedded Systems: ARM Cortex-M (Tiva C), FreeRTOS

Networking: TCP/IP, UDP, socket programming, IPv4 & IPv6 addressing, subnetting, Spanning Tree Protocol (STP)

Databases: MySQL, phpMyAdmin

Developer Tools: Qt6, VS Code, NetBeans, EDAplayground, ModelSim, Keil uVision, MATLAB, Multisim, AutoCAD